# Louis Garrigue

*Postdoctoral researcher in mathematical physics*

*Universität Stuttgart, Germany*
*louis.garrigue@mathematik.uni-stuttgart.de*
*lgarrigue.github.io*

## Positions

Oct 2022 – **Postdoc**, *Stuttgart's university*, in the team of Benjamin Stamm

2020 – 2022 **Postdoc**, *École des ponts, Cermics*, in the team of Éric Cancès

## Education

2017 – 2020 **Doctorate in mathematical quantum physics**, *university Paris-Dauphine*, supervised by Mathieu Lewin, on mathematical aspects of Density functional theory
Referees: Xavier Blanc and Robert Seiringer

2016 – 2017 **Master's degree in fundamental mathematics**, *university Paris-Diderot*

2014 – 2016 **Master's degree in theoretical physics**, *École normale supérieure*

2012 – 2017 **Student and civil servant at the École normale supérieure**

## Preprints

[8] É. Cancès, L. Garrigue, D. Gontier. A simple derivation of moiré-scale continuous models for twisted bilayer graphene (arxiv:2206.05685), preprint (2022).

[7] L. Garrigue. Mixed state representability of entropy-density pairs (arxiv:2203.16441), preprint (2022).

[6] É. Cancès, L. Garrigue, D. Gontier. Second-order homogenization of periodic Schrödinger operators with highly oscillating potentials (arxiv:2112.12008), preprint (2021).

## Publications

[5] L. Garrigue. Building Kohn-Sham potentials for ground and excited states (arxiv:2101.01127), Archive for Rational Mechanics and Analysis (2022).

[4] L. Garrigue. Some properties of the potential-to-ground state map in quantum mechanics (arxiv:2012.04054), Communications in Mathematical Physics (2021).

[3] L. Garrigue. Unique continuation for many-body Schrödinger operators and the Hohenberg-Kohn theorem. II. The Pauli Hamiltonian (arxiv:1901.03207), Documenta Mathematica (2020).

[2] L. Garrigue. Hohenberg-Kohn theorems for interactions, spin and temperature (arxiv:1906.03191), Journal of Statistical Physics (2019).

[1] L. Garrigue. Unique continuation for many-body Schrödinger operators and the Hohenberg-Kohn theorem (arxiv:1804.07564), Mathematical Physics, Analysis and Geometry (2018).

## Contributed talks

Jan 24, 2023 **Seminar PDE and mathematical physics**, *Bordeaux, France*

Dec 13, 2022 **Mathematical physics seminar**, *Stuttgart, Germany*

Dec 8, 2022 **PDE seminar**, *Besançon, France*

Sep 26, 2022 **Conference Aspect'22, Asymptotic Analysis & Spectral Theory**, *Oldenburg, Germany*

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| Aug 17, 2022 | **92nd annual meeting of the International association of applied mathematics and mechanics**, *Aachen, Germany* |
| Jun 21, 2022 | **Workshop at the Center for Advanced Study**, *Norwegian academy of sciences, Oslo, Norway* |
| Apr 11, 2022 | **Seminar of mathematical physics**, *Fourier institute, Grenoble, France* |
| Mar 21, 2022 | **Seminar of mathematical physics**, *Mathematics institute of Bourgogne, Dijon* |
| Mar 7, 2022 | **Seminar of spectral theory**, *Institut Henri Poincaré, Paris* |
| Jan 12, 2022 | **Quantum $N$-body problem GDR, 3rd meeting**, *online* |
| Sep 17, 2021 | **Seminar on mathematical modelling, analysis and simulation of molecular systems (MOANSI)**, *Gamm group, led by Benjamin Stamm, Gero Friesecke and Reinhold Schneider* |
| Aug 6, 2021 | **International congress on mathematical physics**, *Geneva, Switzerland* |
| Jun 14, 2021 | **Seminar on multiscale problems in mathematical physics**, *Angers, France* |
| Mar 31, 2021 | **Seminar of mathematical physics of the Centre for theoretical physics**, *Toulon, France* |
| Mar 3, 2021 | **Seminar of the laboratory of mathematics of the university of Bourgogne**, *Dijon, France* |
| Sep 15, 2020 | **Conference Algorithms in Quantum Molecular Dynamics**, *CIRM, Marseille, France* |
| Sep 15, 2020 | **Seminar of the laboratory**, *Marseille, France* |
| Nov 21, 2019 | **Seminar on Lie theory, geometry and analysis**, *université de Lorraine, Metz, France* |
| Oct 24, 2019 | **The analysis of complex quantum systems**, *CIRM Marseille, France* |
| Sep 22, 2019 | **Density functionals for many-particle systems**, *university of Singapore* |
| Apr 18, 2019 | **PDE seminar of the laboratory of mathematics of Versailles**, *France* |
| Jan 28, 2019 | **Optimal transport methods in Density functional theory**, *Banff research station, Canada* |
| Jul 20, 2018 | **Young researchers symposium of ICMP**, *Montréal, Canada*, satellite event of the International congress of mathematical physics |
| Jul 9, 2018 | **Conference on DFT with magnetic potentials**, *Oslo, Norway* |

## Teaching

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| Oct 2022 – Jan 2023 | **Tutorials of exercices**, *Stuttgart's university*<br>Master's degree course "Numerical methods for eigenvalue problems" |
| Oct 2022 – Jan 2023 | **Students seminar**, *Stuttgart's university*<br>Supervision of two coding and theoretical projects for bachelor's degree students |
| Feb 2022 – Jun 2022 | **Personal projects**, *École des ponts*<br>Presentation and management of a numerical project, coded by the students (first year, equivalent to L3) |
| Sep 2017 – Jun 2020 | **Tutorials of exercices**, *university Paris-Dauphine*<br>Sessions of exercices in analysis (L2), linear algebra (L1) and optimization (M1) |
| Sep 2017 – Jun 2020 | **Physics for all**, *École normale supérieure*<br>Three lectures for the class "Physics for all", addressed to students of the ENS in humanities. Notes at akrajenbrink.weebly.com/teaching.html |
| 2016 – 2017 | **Oral interrogations**, *college Stanislas*, in mathematics, to train students for competitive examinations for entrance in grandes écoles |

## Research interships

Jan – Feb 2016 **Vaccums of a supersymmetric theory**, *laboratory of theoretical physics of the ENS, Paris*, supervised by Jan Troost
Seiberg-Witten theory and solving the structure of inequivalent vaccums of the supersymmetric $\mathcal{N} = 1^*$, $SU(4)$ theory.

Feb – Jul 2015 **Cosmology**, *Berkeley's laboratory of cosmology*, USA, supervised by Robert Cahn
Development and implementation of the algorithm of the assigment of fibers of the telescope to sky objects for the DESI project (dark energy spectroscopic survey), continued and used in the experiment. First light in 2019.

Jun 2013 **Quantum electrodynamics in cavity**, *laboratory Kastler-Brossel ENS, Paris*, supervised by Igor Dotsenko
Sampling quantum measurements after time reversal of a Rydberg atom/photon system in cavity.

## Miscellaneous

July – August 2015 **Internship in a consular scientific service**, *Consulate General of France in San Francisco*, USA, with Philippe Perez
Meetings with researchers of Berkeley laboratories in climate science. Redaction of reports on Californian climatology research for the Cop 21.

2009 **French mathematical olympiad**, *second prize of the region Auvergne*

Reviewer for journals **Communications in Mathematical Physics, Journal of Mathematical Physics**

Coding Julia, Python, C++, Javascript

## Participation to conferences

Jan 9-13, 2023 **The analysis of relativistic quantum system**, *Cirm, Marseille*

Sep 26-30, 2022 **Conference Aspect'22, Asymptotic Analysis & Spectral Theory**, *Oldenburg, Germany*

Sep 21,23, 2022 **Workshop of the Collaborative research center (CRC) on "sparsity and singular structures"**, *Aachen, Germany*

Aug 15-19, 2022 **92$^{\text{nd}}$ annual meeting of the International association of applied mathematics and mechanics**, *Aachen, Germany*

Jun 20-24, 2022 **Workshop at the Center for Advanced Study**, *Oslo, Norway*

Mar 28 - Apr 1, 2022 **Advancing quantum mechanics with mathematics and statistics, workshop 1: multiscale approaches in quantum mechanics**, *IPAM UCLA, Los Angeles*

Feb 23-25, 2022 **Quantum Dynamics GDR, 14$^{\text{th}}$ meeting**, *Toulouse, France*

Jan 10-13, 2022 **Quantum $N$-body problem GDR, 3$^{\text{rd}}$ meeting**, *online*

Sep 16-17, 2021 **Moansi annual meeting**, *Gamm group, led by Benjamin Stamm, Gero Friesecke and Reinhold Schneider*

Aug 25-27, 2021 **Solid Math 2021**, *École des ponts, France*

Oct 27-29, 2021 **Congress of young researchers in applied mathematics**, *École polytechnique, France*

Aug 2-7, 2021 **International congress on mathematical physics and Young researchers symposium**, *Geneva, Switzerland*

Jun 14-16, 2021 **Seminar on multiscale problems in mathematical physics**, *Angers, France*

May 17-28, 2021 **SIAM conference on mathemtical aspects of material science**, *Bilbao, Spain*, online

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| Mar 10-12, 2021 | **Quantum Dynamics GDR, 13$^{\text{th}}$ meeting**, *Dijon, France*, online |
| Feb 8-12, 2021 | **Quantum $N$-body problem GDR, 2$^{\text{nd}}$ meeting**, *online* |
| Sep 14-18, 2020 | **Algorithms in Quantum Molecular Dynamics**, *Cirm, Marseille, France* |
| Sep 2-4, 2020 | **New horizons in density functional theory**, *online*, Faraday discussions workshop |
| Feb 5-7, 2020 | **Quantum Dynamics GDR, 12$^{\text{th}}$ meeting**, *Strasbourg, France* |
| Jan 8-10, 2020 | **Quantum $N$-body problem GDR, 1$^{\text{st}}$ meeting**, *Lille, France* |
| Oct 21-25, 2019 | **The analysis of complex quantum systems : large Coulomb systems and related matters**, *CIRM Marseille, France* |
| Sep 23-27, 2019 | **Density functionals for many-particle systems : mathematical theory and physical applications of effective equations**, *institute for mathematical sciences, university of Singapore* |
| May 27-31, 2019 | **Souriau 2019**, *University Paris-Diderot, France*, Conference in honour of Jean-Marie Souriau |
| Feb 13-15, 2019 | **Quantum Dynamics GDR, 11$^{\text{th}}$ meeting**, *Bordeaux, France* |
| Jan 28- Feb 1$^{\text{st}}$, 2019 | **Optimal transport methods in Density functional theory**, *Banff research station, Canada* |
| Jul 20-28, 2018 | **International congress on mathematical physics and Young researchers symposium**, *Montréal, Canada* |
| Jul 9-13 2018 | **Seminar on DFT with magnetic potentials**, *Oslo, Norway*<br>Entitled "Do electron current densities determine all there is to know ?" |
| Feb 19-24, 2018 | **Mathematical Challenges in Quantum Mechanics**, *Sapienza university, Roma, Italy*, workshop and winter school |
| May 31$^{\text{th}}$-Jun 7$^{\text{th}}$, 2017 | **Density functional theory meets Optimal transport**, *Jyväskylä, Finland*, workshop |